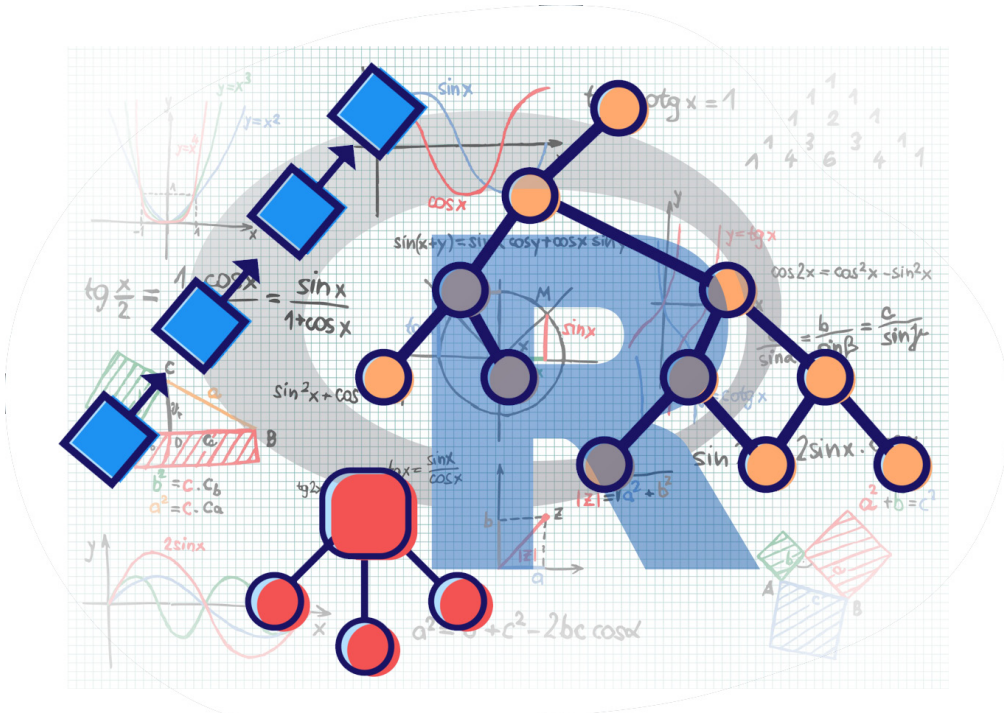


Data Structures in R

In this third article in the 'R, Statistics and Machine Learning' series, we shall explore the various data structures such as vectors, arrays, matrices, lists, factors and data frames that are provided in R to represent data.



We will be using R version 4.1.0 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets. You can just check the version by typing:

```
$ R --version
```

Vector

A vector is a collection that contains a single type of value. It can be numerical, a character, floating point, complex number or have logical values. For example:

```
> c(1, 2, 3, 4)
[1] 1 2 3 4
```

Operations can be performed on vectors. An addition operation is shown below:

```
> c(1, 2, 3, 4) + c(1, 1, 1, 1)
[1] 2 3 4 5
```

If the size of the vectors is not equal in the operation, the smaller sequence is repeated to fill-in, as illustrated below:

```
> c(1, 2, 3, 4) + c(1, 1)
[1] 2 3 4 5
```

```
> c(1, 2, 3, 4) + c(1)
[1] 2 3 4 5
```

You can apply a filter selection, and only those vector members that satisfy the condition are returned. In the following example, only the indices that are divisible by two are listed:

```
> n <- c(1, 2, 3, 4)
> n[n %% 2 == 0]
[1] 2 4
```

The use of negative indices wraps around the vector boundary limits, as shown below:

```
> n[-1:-2]
[1] 3 4
```

An example of character (string) vector in R is as follows: